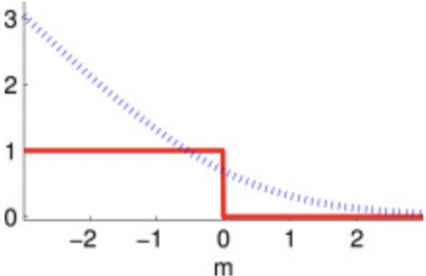
Measures of Similarity/Distance

Procheta Sen



Measures of similarity/distance

General problem

Given two objects $\bar{X}$ and $\bar{Y}$ determine the value of the **similarity** or **distance** between the two objects.

Similarity functions: larger values imply greater similarity

Distance functions: smaller values imply greater similarity
In some domains, such as spatial data, it is more natural to talk about distance functions, whereas in other domains, such as text, it is more natural to talk about similarity functions.
Similarity and distance functions are often expressed in closed form (easy to compute formulas), but in some domains, such as time-series data, they are defined algorithmically and may be computationally expensive.

Measures of similarity/distance for numerical data

Euclidean distance between the vectors $\bar{X}$ and $\bar{Y}$.

$$\mathrm{EucDist}(\bar{X}, \bar{Y}) = \|\bar{X} - \bar{Y}\|_2 = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2} = \sqrt{(\bar{X} - \bar{Y})^T (\bar{X} - \bar{Y})}$$

where $\|\bar{T}\|_2 = \sqrt{\sum_{i=1}^{d} t_i^2}$ denotes the L^2-norm of the vector $\bar{T} = (t_1, t_2, \ldots, t_d)^T$.

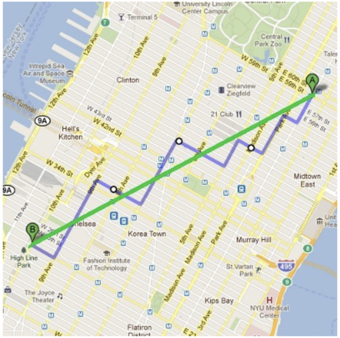
Measures of similarity/distance for numerical data

Manhattan Distance between the vectors $\bar{X}$ and $\bar{Y}$.

$$\mathrm{ManDist}(\bar{X}, \bar{Y}) = \|\bar{X} - \bar{Y}\|_1 = \sum_{i=1}^{d} |x_i - y_i|$$

where $\|\bar{T}\|_1 = \sum_{i=1}^{d} |t_i|$ denotes the L^1-norm of the vector $\bar{T} = (t_1, t_2, \ldots, t_d)^T$.

Example of Manhattan Distance



$EucDist(A, B)$

$ManDist(A, B)$

Figure: Demonstration of Manhattan and Euclidian Distance

Vector norms

- "norm" is a mathematical concept that expresses the "size/length" of a vector
- Popular vector norms in data mining are as follows

$$L^1\text{-norm} \quad \|\bar{X}\|_1 = \sum_{i=1}^{d} |x_i|$$

$$L^2\text{-norm} \quad \|\bar{X}\|_2 = \sqrt{\sum_{i=1}^{d} x_i^2}$$

$$L^0\text{-norm} \quad \|\bar{X}\|_0 = \text{number of non-zero elements in } \bar{X}$$

$$L^\infty\text{-norm} \quad \|\bar{X}\|_\infty = \max\{|x_1|, |x_2|, \ldots, |x_d|\}$$

From vector norms to distances

From a vector norm, one can construct a distance function between pairs of vectors (say, $\bar{X}$ and $\bar{Y}$) as the norm of their difference.

Vector norms

L^1-norm $\|\bar{X}\|_1 = \sum_{i=1}^{d} |x_i|$

L^2-norm $\|\bar{X}\|_2 = \sqrt{\sum_{i=1}^{d} x_i^2}$

L^0-norm $\|\bar{X}\|_0 = \text{no. of non-zero elements in } \bar{X}$

L^∞-norm $\|\bar{X}\|_\infty = \max\{|x_1|, \ldots, |x_d|\}$

Distances from norms

L^1-distance $\|\bar{X} - \bar{Y}\|_1 = \sum_{i=1}^{d} |x_i - y_i|$

L^2-distance $\|\bar{X} - \bar{Y}\|_2 = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$

Measures of similarity/distance for numerical data

Cosine similarity

Let $\bar{X} = (x_1, x_2, \ldots, x_d)$ and $\bar{Y} = (y_1, y_2, \ldots, y_d)$ be vectors in $\mathbb{R}^n$.

$$\mathrm{CosSim}(\bar{X}, \bar{Y}) = \frac{\bar{X}^T \bar{Y}}{\|\bar{X}\|_2 \, \|\bar{Y}\|_2} = \cos(\theta)$$

Where

$$\|\bar{X}\|_2 = \sqrt{\sum_{i=1}^{d} x_i^2}$$

and θ is the angle between the vectors $\bar{X}$ and $\bar{Y}$.

Measuring similarity/distance for numerical data

Cosine similarity

$$\mathrm{CosSim}(\bar{X}, \bar{Y}) = \frac{\bar{X}^T \bar{Y}}{\|\bar{X}\| \, \|\bar{Y}\|} = \cos(\theta)$$

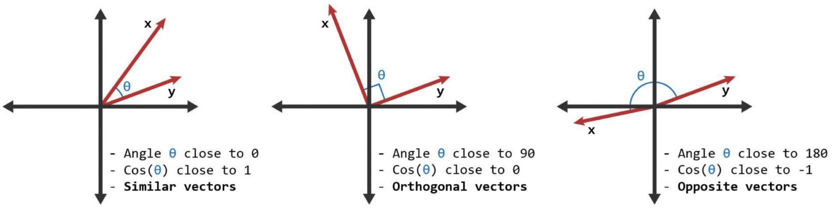


Figure: Demonstration of cosine similarity

Measuring similarity/distance for numerical data

Cosine similarity example (text mining domain):

1. We have a fixed set of **terms (keywords)** of interest
2. Each **term** is assigned a different **dimension**
3. A **document is characterised by a vector** where the value in each dimension corresponds to the frequency of the term appearance in the document
4. **Cosine similarity** then gives a useful measure of how similar two documents are likely to be with respect to their subject matter

Cosine distance

$$\mathrm{CosDist}(\bar{X}, \bar{Y}) = 1 - \mathrm{CosSim}(\bar{X}, \bar{Y})$$

Measures of similarity/distance for set data

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

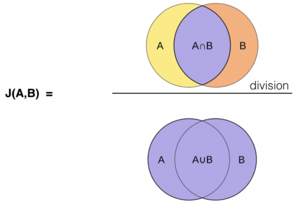


Figure: Diagram of Jaccard Similarity

Jaccard distance: $d_J(A,B) = 1 - J(A,B)$

Measures of similarity/distance for set data

Overlap coefficient:

$$\mathrm{overlap}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

Measures of similarity/distance for binary data

Hamming distance is defined for binary vectors of the same length. It is equal to the number of coordinates where the two vectors differ.

$$\mathrm{Hamming}(\mathbf{X}, \mathbf{Y}) = |\{\, i : x_i \neq y_i \,\}|$$

Example:
$\mathbf{X} = (1, 0, 1, 0, 1, 1)^T$
$\mathbf{Y} = (0, 0, 1, 1, 1, 1)^T$
$\mathrm{Hamming}(\mathbf{X}, \mathbf{Y}) = 2$

Measures of similarity/distance for categorical data

Key differences with respect to numerical data:

- no ordering on the values;
- no natural "difference" operation.

One approach to compute similarity between two objects $\mathbf{X} = (x_1, \ldots, x_d)^T$ and $\mathbf{Y} = (y_1, \ldots, y_d)$ with categorical features is to define:

$$\mathrm{Sim}(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^{d} S(x_i, y_i)$$

where $S(x_i, y_i)$ is the similarity between the feature values x_i and y_i.

Measures of similarity/distance for categorical data

$$\mathrm{Sim}(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^{d} S(x_i, y_i)$$

Specific $S(x_i, y_i)$ (example 1):

$$S(x_i, y_i) = \begin{cases} 1, & \text{if } x_i = y_i \\ 0, & \text{otherwise} \end{cases}$$

Major drawback:

- does not account for the relative frequencies among different feature values;
- e.g., if the value 'Normal' of a fixed feature appears in 99% of the objects and the rest have values 'Cancer' or 'Diabetes', the fact that two objects have this feature equal to 'Normal' tells us less than if they both would have the feature equal to 'Cancer'

Measures of similarity/distance for categorical data

$$\mathrm{Sim}(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^{d} S(x_i, y_i)$$

Specific $S(x_i, y_i)$ (example 2):
Let $p_k(x)$ be the fraction of objects in which the k-th feature takes on the value x in the data set.

$$S(x_i, y_i) = \begin{cases} \dfrac{1}{p_i(x_i)^2}, & \text{if } x_i = y_i \\ 0, & \text{otherwise} \end{cases}$$